ASSIGNMENT NLP – 4 Fine-Tuning BERT on a Kaggle Dataset

BERT Fine-Tuning Text Classification

1**. Introduction**

BERT (Bidirectional Encoder Representations from Transformers) is a transformer-based model developed by Google that understands context in text by reading words in both directions. It is widely used for NLP tasks like sentiment analysis, text classification, question answering, etc.

```
!pip install transformers
!pip install torch
!pip install scikit-learn
!pip install pandas
!pip install numpy
!pip install -U transformers accelerate
```

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```

```
import pandas as pd
```

```
import numpy as np
import torch
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, precision_recall_fscore_support, confusion_matrix
from transformers import AutoTokenizer, AutoModelForSequenceClassification
from transformers import Trainer, TrainingArguments
from transformers import TrainingArguments
```

```
df = pd.read_csv(
    "/content/IMDB Dataset.csv",
    encoding='utf-8',
    engine='python',
    on_bad_lines='skip'
)

df.head()
```

	review	sentiment
0	One of the other reviewers has mentioned that ...	positive
1	A wonderful little production. The...	positive
2	I thought this was a wonderful way to spend ti...	positive
3	Basically there's a family where a little boy ...	negative
4	Petter Mattei's "Love in the Time of Money" is...	positive

Next steps: Generate code with df New interactive sheet

```
df = df.sample(1000, random_state=42)
```

```
import re

def clean_text(text):
    text = text.lower()
    text = re.sub(r'<.*?>', '', text)
    text = re.sub(r'[^a-zA-Z ]', '', text)
    return text

df['review'] = df['review'].apply(clean_text)
df['sentiment'] = df['sentiment'].map({'positive':1, 'negative':0})
```

```
from sklearn.model_selection import train_test_split

train_texts, temp_texts, train_labels, temp_labels = train_test_split(
    df['review'], df['sentiment'], test_size=0.3, random_state=42)

val_texts, test_texts, val_labels, test_labels = train_test_split(
    temp_texts, temp_labels, test_size=0.5, random_state=42)
```

```
from transformers import AutoTokenizer

tokenizer = AutoTokenizer.from_pretrained("bert-base-uncased")

train_encodings = tokenizer(
    list(train_texts),
    truncation=True,
    padding=True,
    max_length=128
)

val_encodings = tokenizer(
    list(val_texts),
    truncation=True,
    padding=True,
    max_length=128
)

test_encodings = tokenizer(
    list(test_texts),
    truncation=True,
    padding=True,
```

```
    max_length=128
)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
```

```
import torch

class Dataset(torch.utils.data.Dataset):
    def __init__(self, encodings, labels):
        self.encodings = encodings
        self.labels = labels.reset_index(drop=True)

    def __getitem__(self, idx):
        item = {key: torch.tensor(val[idx]) for key, val in self.encodings.items()}
        item['labels'] = torch.tensor(self.labels[idx])
        return item

    def __len__(self):
        return len(self.labels)

train_dataset = Dataset(train_encodings, train_labels)
val_dataset = Dataset(val_encodings, val_labels)
test_dataset = Dataset(test_encodings, test_labels)
```

```
from transformers import AutoModelForSequenceClassification

model = AutoModelForSequenceClassification.from_pretrained(
    "bert-base-uncased",
    num_labels=2
)
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster d
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to en
Loading weights: 100%                                                    199/199 [00:00<00:00, 2391.97it/s]
BertForSequenceClassification LOAD REPORT from: bert-base-uncased
Key                                        | Status     |
-------------------------------------------+------------+-
cls.seq_relationship.weight                | UNEXPECTED |
cls.predictions.transform.dense.weight     | UNEXPECTED |
cls.predictions.bias                       | UNEXPECTED |
cls.seq_relationship.bias                  | UNEXPECTED |
cls.predictions.transform.dense.bias       | UNEXPECTED |
cls.predictions.transform.LayerNorm.weight | UNEXPECTED |
cls.predictions.transform.LayerNorm.bias   | UNEXPECTED |
classifier.weight                          | MISSING    |
classifier.bias                            | MISSING    |

Notes:
- UNEXPECTED:   can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING:  those params were newly initialized because missing from the checkpoint. Consider training on your downstream task.
```

```
from sklearn.metrics import accuracy_score, precision_recall_fscore_support

def compute_metrics(pred):
    labels = pred.label_ids
    preds = pred.predictions.argmax(-1)

    precision, recall, f1, _ = precision_recall_fscore_support(labels, preds, average='binary')
    acc = accuracy_score(labels, preds)

    return {
        'accuracy': acc,
        'precision': precision,
        'recall': recall,
        'f1': f1
    }
```

```
from transformers import TrainingArguments
```

```
training_args = TrainingArguments(
    output_dir="./results",
    num_train_epochs=1,
    per_device_train_batch_size=4,
    per_device_eval_batch_size=4,
    learning_rate=2e-5,
    logging_steps=200
)
```

```
from transformers import Trainer

trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=train_dataset,
    eval_dataset=val_dataset,
    compute_metrics=compute_metrics
)
```

```
trainer.train()
```

```
/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memory' argument is set as true bu
  super().__init__(loader)
```

[175/175 15:53, Epoch 1/1]

Step	Training Loss

Writing model shards: 100% 1/1 [00:05<00:00, 5.29s/it]

```
TrainOutput(global_step=175, training_loss=0.626141836983817, metrics={'train_runtime': 974.4363, 'train_samples_per_second':
0.718, 'train_steps_per_second': 0.18, 'total_flos': 46044434688000.0, 'train_loss': 0.626141836983817, 'epoch': 1.0})
```

```
predictions = trainer.predict(val_dataset)
```

```
/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memory' argument is set as true bu
  super().__init__(loader)
```

```
from sklearn.metrics import accuracy_score, precision_recall_fscore_support

preds = predictions.predictions.argmax(-1)
labels = predictions.label_ids

precision, recall, f1, _ = precision_recall_fscore_support(labels, preds, average='binary')
accuracy = accuracy_score(labels, preds)

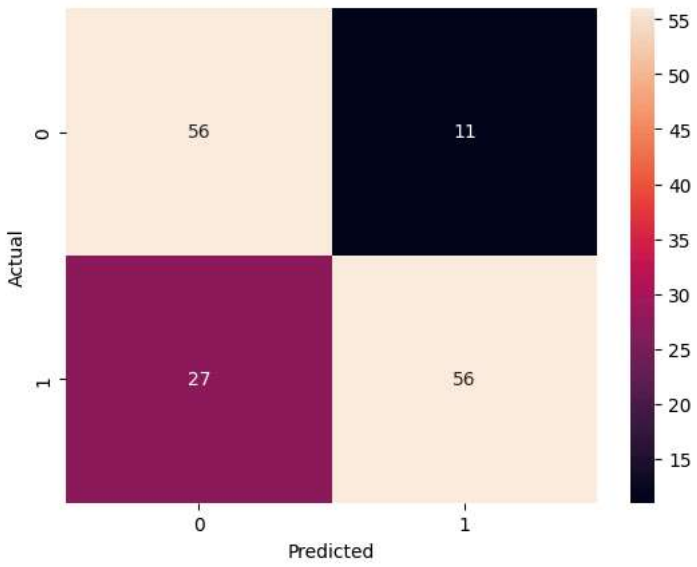
print("Accuracy:", accuracy)
print("Precision:", precision)
print("Recall:", recall)
print("F1 Score:", f1)
```

```
Accuracy: 0.7466666666666667
Precision: 0.835820895522388
Recall: 0.6746987951807228
F1 Score: 0.7466666666666667
```

```
from sklearn.metrics import confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt

cm = confusion_matrix(labels, preds)

sns.heatmap(cm, annot=True, fmt='d')
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```



Start coding or generate with AI.